UNDERSTANDING REGIONAL LANGUAGE IN LARGE LANGUAGE MODELS: A SHAP-BASED ANALYSIS OF CONTEXT DEPENDENCE

Guido Ivetta[1,2], Sofía Martinelli[1,2,3]
[1]Universidad Nacional de Córdoba (Argentina), [2]Fundación Vía Libre, [3]CONICET

ABSTRACT

Large Language Models (LLMs) often exhibit lower performance on underrepresented language varieties. While performance-based evaluations show how well models interpret these varieties, they do not necessarily capture how different components of the input influence their predictions. In this work, we introduce a community-collected dataset of Argentine expressions and compare model performance on these expressions with U.S. slang, evaluating Gemma 4B with and without usage examples. We study both how accurately the model defines these expressions, using semantic similarity, and how much the target expression and its usage context contribute to model performance, using SHAP (Lundberg & Lee, 2017). Our results show that the model performs worse on Argentine expressions than on U.S. slang in the absence of context. Providing usage examples improves performance in both cases, with a larger relative gain for Argentine expressions (19.4% vs. 14.1%). SHAP analysis further shows a greater relative contribution of contextual examples for Argentine expressions. Our findings illustrate how explainability methods can complement evaluations of LLMs.

1 INTRODUCTION

Linguistic diversity remains underrepresented in current Large Language Models (LLMs). Although these systems are trained on massive multilingual corpora, the distribution of linguistic resources is highly imbalanced: only a small group of "winner" languages and dominant language varieties receives disproportionate attention in data collection and model development, while the vast majority of languages and regional varieties remain underrepresented (Joshi et al., 2020). Consequently, LLMs exhibit lower understanding of non-standard varieties and often generate responses perceived as more stereotypical, condescending, and degrading than those produced for standard varieties (Fleisig et al., 2024). In this context, the harms caused by LLMs can disproportionately affect already marginalized communities by reinforcing existing social prejudices (Ducel et al., 2026).

Even in widely spoken languages, LLMs often perpetuate a homogenized version of the language. Spanish illustrates this phenomenon well, as it encompasses a diverse set of diatopic varieties that current models frequently fail to distinguish. Previous work has shown that models tend to favor dominant varieties, such as Peninsular Spanish, leading to weaker recognition of linguistic features associated with other Spanish varieties, particularly those spoken across Latin America (Mayor-Rocher et al., 2025). Among these, Argentine regionalisms remain unexplored in LLM evaluation.

Existing evaluations of regional languages have focused on measuring model performance across different Natural Language Processing (NLP) tasks, providing limited insight into the mechanisms underlying model errors. For example, Attia et al. (2026) evaluate LLMs on Arabic figurative language and show that models achieve lower accuracy on multiple-choice comprehension tasks when interpreting Egyptian Arabic idioms without additional context. Although providing a usage example improves task performance, the study does not examine how contextual information contributes to the model's interpretation of the expression. Therefore, it remains unclear how much models rely on their internal linguistic knowledge versus contextual information when interpreting culturally grounded expressions.

Our work addresses this question by combining accuracy evaluation with SHAP-based attribution analysis (Lundberg & Lee, 2017). We compare two linguistic contexts: regional expressions from Argentina and slang from the United States. The Argentine dataset consists of a community-collected benchmark gathered with more than 800 students from FAMAF (UNC) and more than 1,500 secondary school teachers from the province of Córdoba through the EDIA tool (Maina et al., 2024). American slang is represented by the OpenSub-Slang (Sun et al., 2024) benchmark. We hypothesize that LLMs will correctly define American slang using their internal linguistic knowledge, whereas they will struggle with Argentine regionalisms unless contextual information is provided. Through SHAP-based explanations, we further investigate whether models rely more heavily on contextual information when interpreting Argentine regionalisms than when interpreting American slang.

Specifically, we seek to answer the following research questions:

- **RQ1:** Do LLMs correctly define American slang even without additional context, while systematically failing on Argentine regionalisms under the same condition?
- **RQ2:** Which parts of the input contribute most to the model's response when interpreting culturally or regionally grounded expressions?
- **RQ3:** Does adding a usage example significantly increase accuracy for Argentine regionalisms, but not for American slang?

We propose the following hypotheses:

- **H1:** LMs will achieve high accuracy when defining American slang without context, while showing significantly lower accuracy for Argentine regionalisms under the same condition.
- **H2:** SHAP explanations will show that the usage example receives a significantly higher attribution weight in Argentine expressions than in US ones.
- **H3:** Adding a usage example will substantially improve accuracy for Argentine regionalisms, while producing only a marginal improvement for American slang.

2 Related Work

LLMs Evaluation Across Language Varieties. Current LLMs are predominantly trained and aligned on English and a small number of dominant language varieties (Joshi et al., 2020). As a result, recent evaluations consistently report lower performance on non-standard language varieties. At a global scale, Faisal et al. (2024) evaluated 281 varieties across 10 diverse NLP tasks, revealing widespread performance disparities not only when transferring from Standard English to other varieties, but also between dominant standard languages and their respective regional dialects. This pattern has been observed across multiple languages. For example, LLMs perform substantially worse on regional Arabic dialects than on Modern Standard Arabic, particularly for low-resource varieties (Mousi et al., 2025), and they exhibit significant degradation when processing non-standard English varieties such as African American Vernacular English (Lin et al., 2025). Similar findings have been reported for Spanish, where models tend to favor Peninsular Spanish over other regional varieties (Martínez et al., 2025). However, existing evaluations of Argentine Spanish have focused exclusively on Rioplatense Spanish (Martínez et al., 2025), while broader multilingual benchmarks such as IberBench (Ángel González et al., 2025) evaluate several Spanish varieties—including Mexican, Uruguayan, and Peruvian Spanish—but omit Argentine regional varieties. Our work addresses this gap by introducing a community-collected dataset of Argentine regionalisms to study the performance of LLMs, following the methodology of Ivetta et al. (2026; 2025).

While these studies highlight performance disparities, they rely on standard metrics such as accuracy, F1-score, or BLEU, offering limited insight into model error mechanisms. For example, Attia et al. (2026) show that LLMs struggle to interpret colloquial Egyptian idioms in multiple-choice tasks without context. Although adding a usage example improves accuracy, the authors do not evaluate whether this gain reflects a systematic reliance on contextual cues. Thus, current evaluations show that models fail on underrepresented languages but lack explainability methods to uncover how different parts of the input drive the output. We address this gap by pairing performance evaluation with SHAP feature attribution (Lundberg & Lee, 2017) to uncover the mechanisms behind these disparities.

Model Explainability. SHAP (SHapley Additive exPlanations) (Lundberg & Lee, 2017) was introduced as a unified framework for interpreting the predictions of complex models. The method is grounded in cooperative game theory: each input feature is treated as a *player* in a game whose *payout* is the model's prediction, and Shapley values (Shapley, 1952) are used to fairly distribute that payout among the features based on their marginal contribution across all possible feature subsets. In particular, Kernel SHAP is a model-agnostic approximation that treats the model as a black box and only requires a function mapping inputs to a scalar output. Although SHAP was originally developed and evaluated on classification and regression models, Kernel SHAP made it applicable regardless of the underlying model architecture or task type. In this work, we leverage this model-agnostic property of Kernel SHAP to analyze a generative LLM. We define two input components—a regional expression or slang and its usage example—and use the semantic similarity between the model's generated definition and a reference definition as the scalar function required by Kernel SHAP. This adaptation allows us to quantify the relative contribution of each component to the model's output without requiring access to internal model weights or gradients.

3 Experimental Setup

This study examines how LLMs interpret situated language and how this is shaped by the availability of contextual information, comparing American slang and Argentine regional expressions. We combine accuracy-based evaluation with post-hoc explanations to assess both model performance and the contribution of context to predictions. This section describes the datasets, model setup, experimental conditions, and evaluation procedures.

Origin	Expression	Definition	Example
ARG	La gorra	La policía	Corré que viene la gorra
	No le llega agua al tanque	Persona de comprensión lenta	No entendés nada, no te llega agua al tanque
	Pritiado	Mezcla de vino tinto y pritty	Vamos a tomar un pritiado
	Me hice aca	Golpearse o lastimarse involuntariamente	Estaba corriendo, se me cruzó un perro y me hice aca
	Pipí cucú	Perfecto, impecable	El arreglo del coche quedó pipí cucú
US	Weirdo	Strange person	No wonder he's a complete weirdo.
	Bogs	Toilets, lavatories	Somebody told me it's four miles to the bogs.
	Nailed it	To prevail, win or triumph	You nailed it today, man, but you always do.
	Keep your hair on	Calm down	All right, Father, keep your hair on!
	Chum	A close friend, a room-mate	Chum, come and pick me up.

Table 1: Sample entries from the Argentine Spanish expressions and US English slang benchmarks.

3.1 Datasets

We selected two benchmarks: Argentine Spanish regional expressions and American English slang, each comprising a target expression, its reference definition, and an illustrative example sentence. Sample entries are shown in Table 1.

U.S. Slang. This context is represented by two datasets containing English slang expressions from the United States. The first is **OpenSub-Slang** (Sun et al., 2024), a benchmark derived from movie subtitles that includes 7,488 human-annotated slang instances.

Argentine Regional Expressions. The Argentine dataset contains 5,254 regional expressions contributed by speakers from different provinces of Argentina, with the majority of contributions originating from Córdoba. The corpus was collected through the EDIA annotation platform, following the community-based methodology validated in the *HESEIA* (Ivetta et al., 2025) and *LACES* (Ivetta et al., 2026) projects. Data collection involved more than 800 first-year students from FAMAF (Universidad Nacional de Córdoba) who provided expressions, definitions, and examples from their home provinces. The dataset was later expanded with contributions from over 1,500 secondary school teachers from Córdoba through a professional development program recognized by the Ministerio de Educación of Córdoba.

3.2 MODEL

We employed the *Gemma 4B* model (Gemma Team & DeepMind, 2026), leveraging the computational infrastructure provided by the Centro de Computación de Alto Desempeño (CCAD)[1] at the Universidad Nacional de Córdoba. Since the model is accessed only through generated text outputs, it is treated as a black-box system. This setting motivates the use of model-agnostic explanation methods to analyze the factors that influence the model's interpretations. We selected *Gemma 4B* since it provides a favorable trade-off between performance and computational cost, which is particularly important given the large number of model evaluations required by the SHAP analysis.

3.3 PROMPTS

The experiment evaluates the effect of contextual information on the interpretation of regional expressions and slang. Each expression is evaluated using two prompts in its original language (English or Spanish).

Without context. Only the target expression is provided to the model.

```
What does the following expression mean?
Expression: {phrase}
Respond with a brief definition.
```

With context. The target expression is accompanied by an example sentence illustrating its use.

```
What does the following expression mean?
Expression: {phrase}
Usage example: {example}
Respond with a brief definition.
```

Comparing these two conditions allows us to measure the extent to which the model relies on contextual information when interpreting non-standard language.

3.4 EVALUATION

The generated definitions are analyzed from two complementary perspectives. First, we evaluate whether the model produces correct interpretations. Second, we investigate which parts of the input contribute most to the model's performance.

3.4.1 GENERATED DEFINITIONS

To evaluate the accuracy of the generated definitions, we use cosine similarity as a proxy. Specifically, we compute the cosine similarity between each generated definition and its corresponding reference definition using multilingual sentence embeddings *paraphrase-multilingual-MiniLM-L12-v2* (Reimers & Gurevych, 2019). Higher similarity values indicate greater semantic alignment with the intended meaning of the expression and, consequently, higher definition accuracy. We compare the average cosine similarity across expressions for Argentine expressions and U.S. slang, both with and without usage examples. We expect U.S. slang to achieve higher accuracy than Argentine

[1] `https://supercomputo.unc.edu.ar/`

expressions in the absence of context, while Argentine expressions are expected to show a larger improvement when contextual information is provided.

3.4.2 SHAP-Based Analysis

While accuracy measures the quality of the generated definitions, it does not reveal how much the model relies on its internal linguistic knowledge versus the contextual information contained in the example. To investigate this aspect, the evaluation is complemented by SHAP-based explanations. Each input instance is represented by two features: the target expression and its corresponding usage example. Since the model produces free-text definitions, the outputs are first converted into a numerical score by computing the cosine similarity between the generated definition and the dataset's reference definition. This similarity score serves as the target variable explained by SHAP. We employ Kernel SHAP using a subset of the input instances as a background dataset. During the attribution process, masked features are not removed from the input. Instead, they are replaced with the corresponding feature values from the background dataset. The resulting SHAP values estimate the contribution of both the target expression and the usage example to the final similarity score. This analysis allows us to compare the influence of contextual information across linguistic groups and to test whether the usage example receives a significantly higher attribution weight for Argentine regional expressions than for U.S. slang, as postulated in *H2*.

4 Results

4.1 Cosine Similarity and Definition Accuracy

To evaluate the baseline comprehension of our target model, we first examine the generated definitions in the zero-context condition. As shown in Figure 1, *Gemma 4B* exhibits a performance disparity between the two linguistic varieties when no usage example is provided. For U.S. slang, the model achieves a relatively high baseline semantic alignment with the reference definitions. Conversely, performance on Argentine regional expressions (ARG) is lower under the exact same zero-context condition. This systematic degradation confirms our first hypothesis (**H1**) and aligns with prior findings that current LLMs rely on dominant language representations, struggling to retrieve accurate semantic representations for underrepresented sociolects.

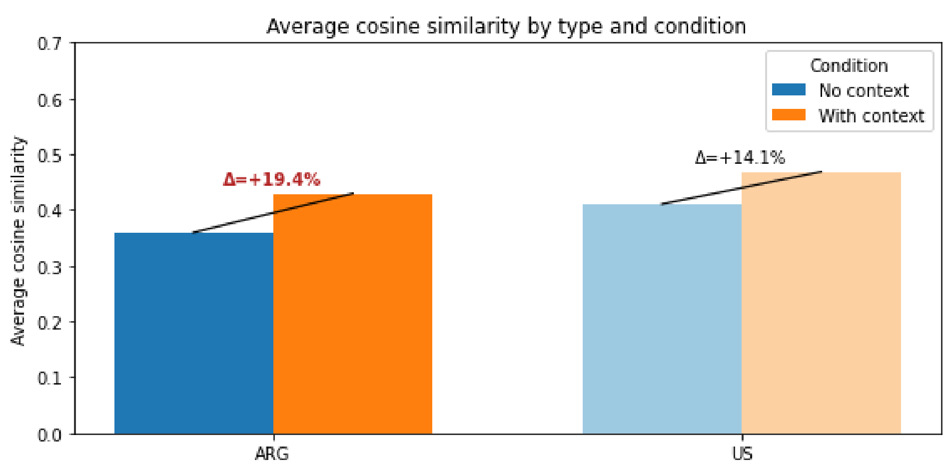

Figure 1: Average cosine similarity between model-generated definitions and gold references by linguistic variety and evaluation condition. Scores are compared for Argentine regionalisms (ARG) and U.S. slang (US) across zero-context and with-context settings. While both varieties benefit from contextual information, Argentine regionalisms exhibit a substantially larger relative gain (19.4% vs. 14.1%), illustrating the model's heightened context-dependence when processing underrepresented regional dialects.

Introducing a usage example into the prompt consistently improves the mean cosine similarity across both datasets; however, the magnitude of this improvement varies considerably by region, directly addressing **RQ1** and **RQ3**. Specifically, adding contextual information yields a 19.4% relative increase in semantic similarity for Argentine regionalisms, compared to only a 14.1% increase for U.S. slang. This differential gain provides empirical support for **H3**: while the model is largely capable of defining U.S. slang using its internal linguistic knowledge—benefiting only moderately from contextual disambiguation—it is highly dependent on situational context to successfully interpret Argentine regionalisms. Without the illustrative example, the model frequently fails to bridge the cultural and linguistic gap inherent in the Argentine benchmark.

4.2 SHAP Attribution Analysis

To address **RQ2**, we analyze the feature attribution weights assigned by Kernel SHAP to the two primary input components: the target expression and the usage example. This evaluation measures the degree to which the model relies on the provided context versus its internal linguistic knowledge when generating definitions.

As shown in Figure 2, the usage example receives a higher average absolute SHAP value than the target phrase across both linguistic varieties. However, the relative difference between feature contributions varies between the two datasets, supporting **H2**.

For U.S. slang, the usage example shows a 28.8% relative increase in attribution weight over the target phrase. This distribution indicates that while the example contributes to the output, the model also utilizes the target phrase itself, suggesting it retrieves existing representations of U.S. slang from its parametric memory.

For Argentine regionalisms, reliance on the usage example is higher, exhibiting a 53.0% relative increase in average SHAP attribution compared to the target phrase. Without accurate prior representations of the regional meaning, the model assigns lower attribution to the phrase and relies primarily on the context provided by the example sentence.

These attribution patterns explain the accuracy differences observed in Section 4.1. For dominant language varieties, the model utilizes both internal lexical knowledge and context. For underrepresented regional varieties, accurate interpretation depends primarily on the surrounding context.

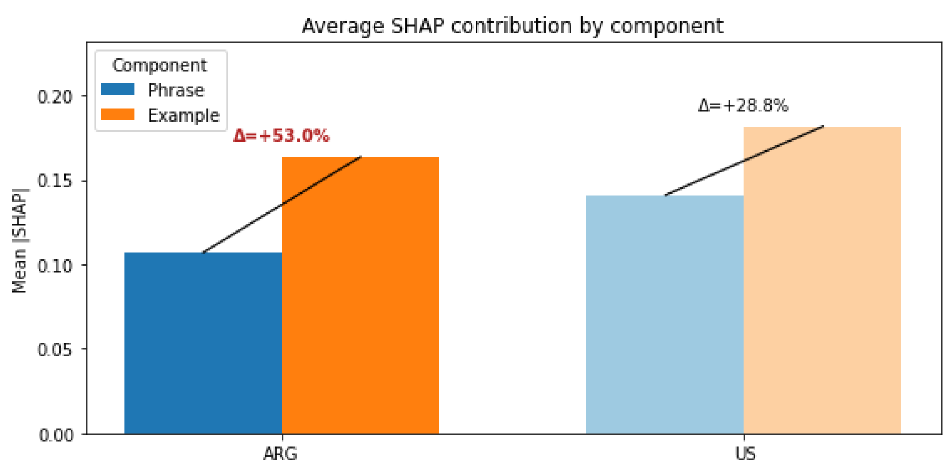

Figure 2: Average absolute SHAP contribution by input component for Argentine regionalisms (ARG) and U.S. slang (US). Bars show the mean SHAP value for the target phrase and the contextual example, and annotations indicate the relative change from phrase to example. In both groups, the example contributes more than the phrase. The model shows higher context dependence for ARG (+53.0%) than for US (+28.8%).

5 Conclusion

In this work, we investigated how an LLM interprets Argentine expressions compared with U.S. slang, combining accuracy evaluation with SHAP-based explanations. Our results support our three hypotheses: Gemma 4B achieves lower definition accuracy for Argentine expressions in the absence of context (H1), benefits more from the addition of usage examples for these expressions (H3), and shows greater reliance on contextual examples when interpreting them (H2). Overall, our findings suggest that the lower performance observed for Argentine expressions is accompanied by greater dependence on contextual information. Our work illustrates how explainability methods can complement performance-based evaluation to provide a better understanding of linguistic disparities in LLMs.

6 Limitations

Our findings should be interpreted in light of the following limitations. First, the comparison between American slang and Argentine regionalisms may not be entirely fair, as the two benchmarks differ in construction methodology and may vary in data quality or intrinsic difficulty. This should be kept in mind when interpreting differences between the two benchmarks. Second, due to the computational cost of the SHAP-based analysis, our experiments are limited to a single model (*Gemma 4B*), and the findings may not generalize to models of different scales or architectures. Third, our dataset is not uniformly representative of Argentina: most contributions come from Cordoba, so our findings may reflect patterns specific to that province. We similarly do not examine regional variation within the United States, as no benchmark distinguishing within-country slang varieties was available.

Acknowledgments

The authors would like to acknowledge the computational resources provided by the Centro de Computación de Alto Desempeño (CCAD) at the Universidad Nacional de Córdoba (https://ccad.unc.edu.ar/), which is part of the Sistema Nacional de Computación de Alto Desempeño (SNCAD), MinCyT, Argentina. Furthermore, we are deeply grateful to the more than 800 first-year students from FAMAF (UNC) and the over 1,500 secondary school teachers from the province of Córdoba whose valuable contributions made the construction of the Argentine regional expressions dataset possible.

References

Mena Attia, Aashiq Muhamed, Mai Alkhamissi, Thamar Solorio, and Mona T. Diab. Beyond understanding: Evaluating the pragmatic gap in LLMs' cultural processing of figurative language. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7238–7265, Rabat, Morocco, March 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.eacl-long.341.

Fanny Ducel, Aurélie Névéol, Vidit Khazanchi, Loïc Leclere, and Arthur Pedrini. Code-switching as a bias indicator in llms: "the consequences are not the same para nosotros". In *Proceedings of the 15th Biennial Language Resources and Evaluation Conference (LREC 2026)*, Palma de Mallorca, Spain, May 2026. doi: 10.63317/2mq6kqjk9bng. hal-05529786.

Fahim Faisal, Orevaoghene Ahia, Aarohi Srivastava, Kabir Ahuja, David Chiang, Yulia Tsvetkov, and Antonios Anastasopoulos. DIALECTBENCH: An NLP benchmark for dialects, varieties, and closely-related languages. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 14412–14454, Bangkok, Thailand, August 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.777.

Eve Fleisig, Genevieve Smith, Madeline Bossi, Ishita Rustagi, Xavier Yin, and Dan Klein. Linguistic bias in ChatGPT: Language models reinforce dialect discrimination. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pp. 13541–13564, Miami, Florida, USA, November 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.emnlp-main.750.

Gemma Team and Google DeepMind. Gemma 4 technical report, 2026. URL `https://arxiv.org/abs/2607.02770`.

Guido Ivetta, Marcos J Gomez, Sofía Martinelli, Pietro Palombini, M Emilia Echeveste, Nair Carolina Mazzeo, Beatriz Busaniche, and Luciana Benotti. HESEIA: A community-based dataset for evaluating social biases in large language models, co-designed in real school settings in Latin America. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 25095–25117, Suzhou, China, November 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.1275.

Guido Ivetta, Pietro Palombini, Sofía Martinelli, Marcos J Gomez, M Emilia Echeveste, Sunipa Dev, Vinodkumar Prabhakaran, and Luciana Benotti. Adaptive data collection for Latin-American community-sourced evaluation of stereotypes (LACES). In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 4177–4190, San Diego, California, United States, July 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-acl.203.

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. The state and fate of linguistic diversity and inclusion in the NLP world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 6282–6293, Online, July 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.560.

Fangru Lin, Shaoguang Mao, Emanuele La Malfa, Valentin Hofmann, Adrian de Wynter, Xun Wang, Si-Qing Chen, Michael J. Wooldridge, Janet B. Pierrehumbert, and Furu Wei. Assessing dialect fairness and robustness of large language models in reasoning tasks. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 6317–6342, Vienna, Austria, July 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.317.

Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Proceedings of the 31st International Conference on Neural Information Processing Systems*, NIPS'17, pp. 4768–4777, Red Hook, NY, USA, 2017. Curran Associates Inc. ISBN 9781510860964.

Hernán Maina, Laura Alonso Alemany, Guido Ivetta, Mariela Rajngewerc, Beatriz Busaniche, and Luciana Benotti. Exploring stereotypes and biases in language technologies in latin america. *Commun. ACM*, 67(8):54–56, August 2024. doi: 10.1145/3653322.

Gonzalo Martínez, Marina Mayor-Rocher, Cris Pozo Huertas, Nina Melero, María Grandury, and Pedro Reviriego. Spanish is not just one: A dataset of spanish dialect recognition for llms. *Data in Brief*, 63, 09 2025. doi: 10.1016/j.dib.2025.112088.

Marina Mayor-Rocher, Cristina Pozo, Nina Melero, Gonzalo Martínez, María Grandury, and Pedro Reviriego. It's the same but not the same: Do llms distinguish spanish varieties?, 2025. URL `https://arxiv.org/abs/2504.20049`.

Basel Mousi, Nadir Durrani, Fatema Ahmad, Md. Arid Hasan, Maram Hasanain, Tameem Kabbani, Fahim Dalvi, Shammur Absar Chowdhury, and Firoj Alam. AraDiCE: Benchmarks for dialectal and cultural capabilities in LLMs. In *Proceedings of the 31st International Conference on Computational Linguistics*, pp. 4186–4218, Abu Dhabi, UAE, January 2025. Association for Computational Linguistics.

Nils Reimers and Iryna Gurevych. Sentence-bert: Sentence embeddings using siamese bert-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 11 2019.

Lloyd S. Shapley. *A Value for N-Person Games*. RAND Corporation, Santa Monica, CA, 1952. doi: 10.7249/P0295.

Zhewei Sun, Qian Hu, Rahul Gupta, Richard Zemel, and Yang Xu. Toward informal language processing: Knowledge of slang in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 1683–1701, Mexico City, Mexico, June 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.94.

José Ángel González, Ian Borrego Obrador, Álvaro Romo Herrero, Areg Mikael Sarvazyan, Mara Chinea-Ríos, Angelo Basile, and Marc Franco-Salvador. Iberbench: Llm evaluation on iberian languages, 2025. URL https://arxiv.org/abs/2504.16921.